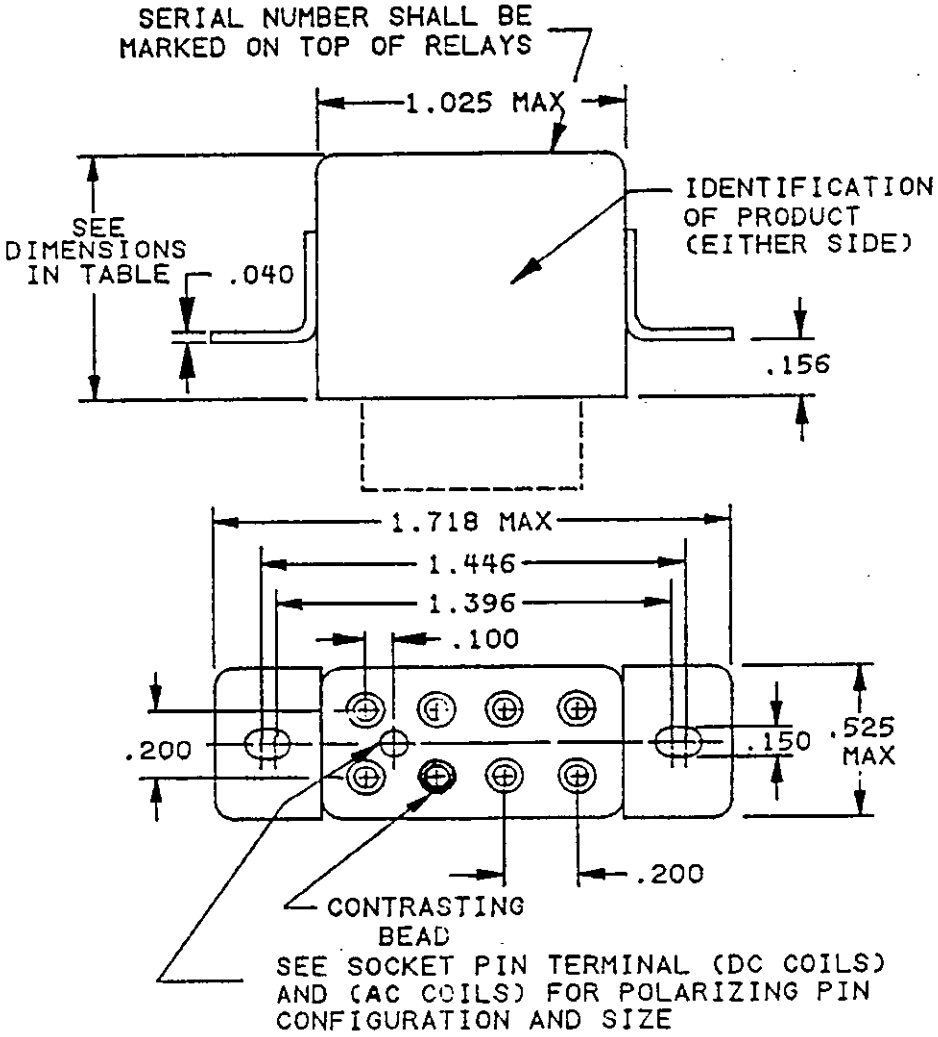
MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC (TYPE I ER), ESTABLISHED RELIABILITY, PERMANENT MAGNET DRIVE, 10 AMPERES, 2PDT, ALL WELDED, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 24 OCTOBER 1986.
FOR NEW DESIGN USE MS27401 (SEE TABLE I HEREIN).

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-6106.

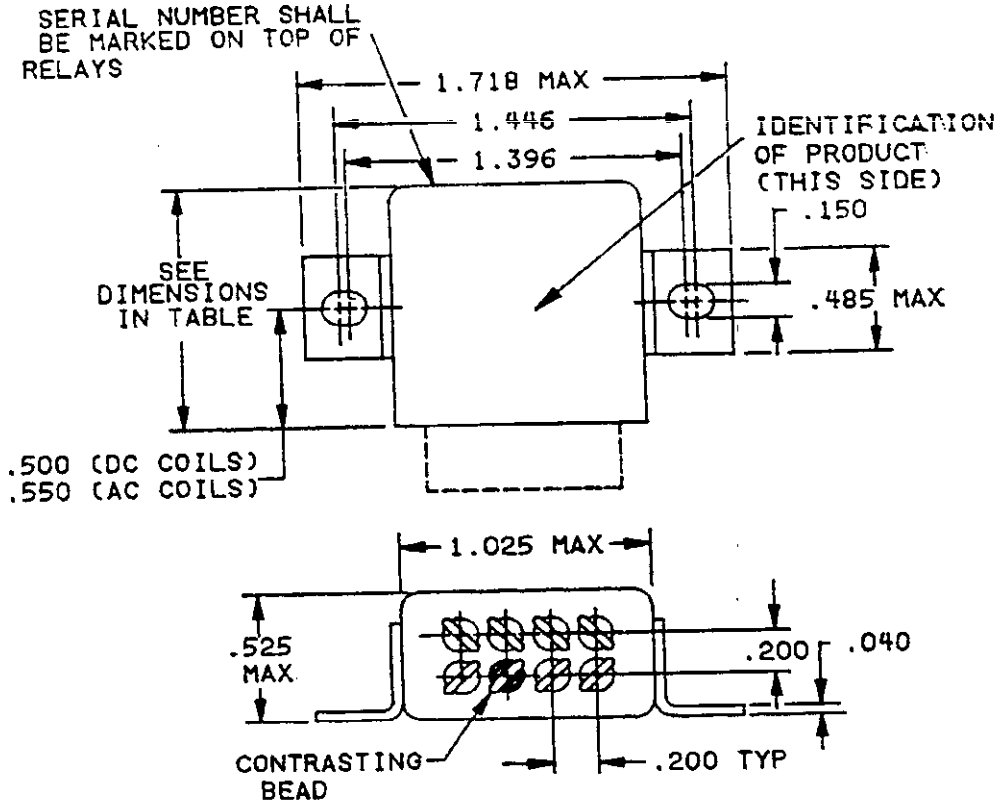


RAISED VERTICAL FLANGE MOUNT (RVFM)

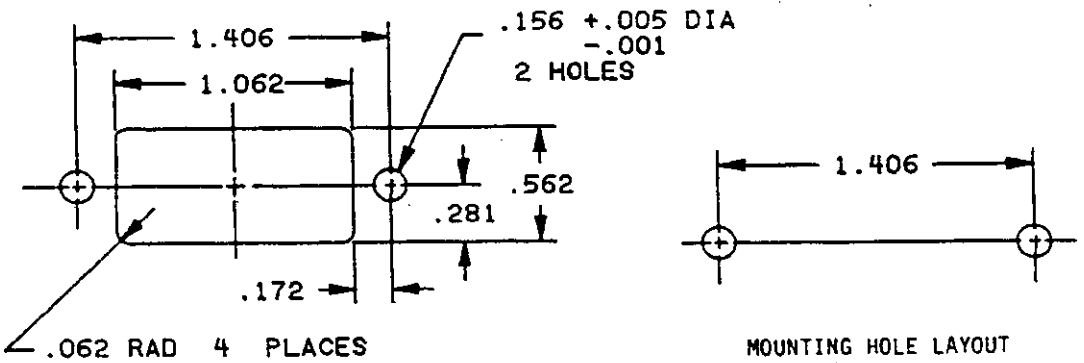
FIGURE 1. Dimensions and configurations.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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HORIZONTAL FLANGE MOUNT (HFM)

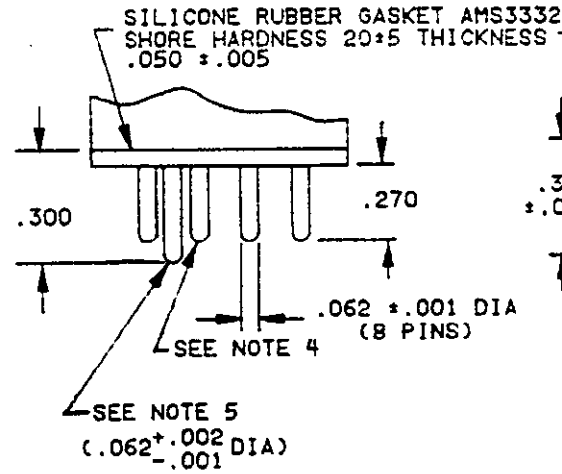


MOUNTING HOLE LAYOUT FOR RVFM (SOLDER HOOK TERMINALS)

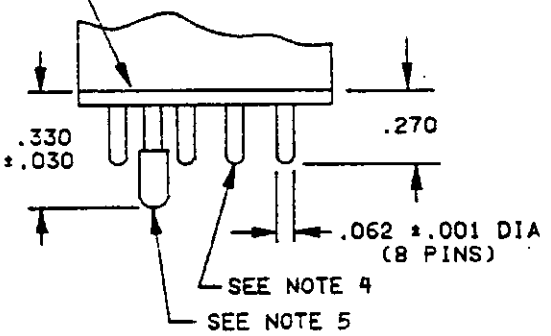
MOUNTING HOLE LAYOUT FOR HFM

FIGURE 1. Dimensions and configurations - Continued.

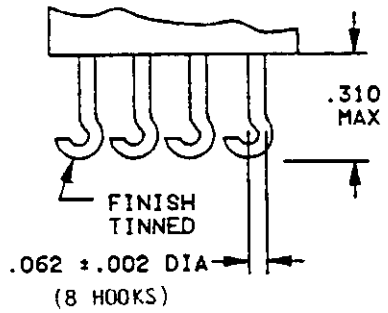
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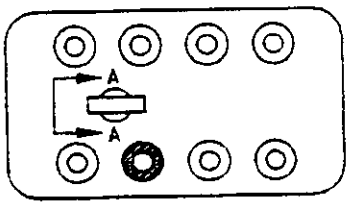
SOCKET-PIN TERMINAL (DC COILS)



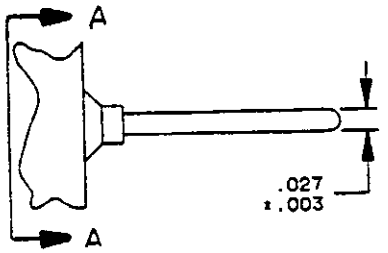
SOCKET-PIN TERMINALS (AC COILS)

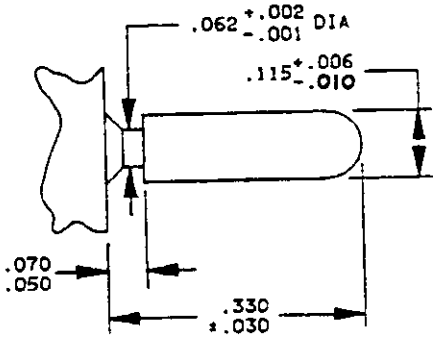


SOLDER-HOOK TERMINAL

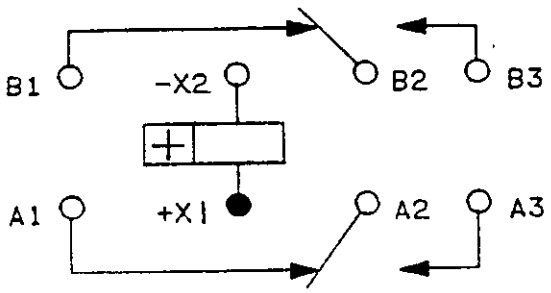


TERMINAL LAYOUT FOR AC COILS





POLARIZING PIN DETAIL FOR AC COILS
(SEE NOTES 4 AND 5)



CIRCUIT DIAGRAM (TERMINAL VIEW)
(POLARIZED MONOSTABLE)
(SEE NOTES 7, 8, AND 9)

FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm
.001	0.03	.172	4.37
.002	0.05	.200	5.08
.003	0.08	.270	6.86
.005	0.13	.281	7.14
.006	0.15	.300	7.62
.010	0.25	.310	7.87
.027	0.69	.330	8.38
.030	0.76	.465	11.81
.040	1.02	.500	12.70
.050	1.27	.525	13.34
.060	1.52	.562	14.27
.062	1.57	1.025	26.04
.070	1.78	1.062	26.97
.100	2.54	1.396	35.46
.115	2.92	1.406	35.71
.150	3.81	1.446	36.73
.156	3.96	1.718	43.64

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm) for three place decimals.
3. Metric equivalents are given for general information only.
4. Finish shall be gold plate per MIL-G-45204, type II, class I. Underplating, nickel (50 to 150 microinches thick minimum).
5. Finish shall be gold plate or tin plate.
6. Terminal numbers do not appear on the relay header provided. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
7. Coil polarity not applicable to ac versions -003, -004, -007, -008, -010, and -011.
8. DC versions of this relay shall not operate or be damaged by reverse polarity. Semiconductors shall not be used for this purpose.
9. Permanent magnet drive consists of permanent magnet with its flux path switched and combined with the electro-magnetic flux.
10. The degassing of these relays shall be performed without the enclosure sealed to the header. Relay shall be constructed without a seal-off (tip-off) hole. When relay starts through degassing process it shall not be exposed to the atmosphere until final sealing is completed (see degassing procedure for MIL-R-6106).

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Weight:

M6106/7-001, -002, -007, -009, -010, -011 and -012: 0.09 pound (41.02 grams (g)) maximum.

M6106/7-003, -004, -005, -006, and -008: 0.10 pound (45.36 grams) maximum.

Temperature range: -70°C to +125°C.

Maximum altitude rating: 300,000 feet.

Shock G-level: 200 g.

Duration: 6 milliseconds (ms).

Maximum duration contact opening 1/: 10 microseconds (µs).

Vibration (sinusoidal):

G-level: 30 g.

Frequency range: 10 to 3,000 Hz.

Vibration (random):

Applicable standard: MIL-STD-202, method 214, test condition IG.

Power spectral density: 0.4 G^2/Hz.

Frequency range: 50 to 2,000 Hz.

Acceleration: 15 g.

Insulation resistance:

Initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric withstanding voltage (sea level):

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Auxiliary contacts	---	---
All other points	1,250 V rms	1,000 V rms

Dielectric withstanding voltage (altitude): 2/

	Initial	After life tests
	80,000 feet	300,000 feet
Coil to case	350 V rms	500 V rms
Auxiliary contacts	---	---
All other points	350 V rms	500 V rms

Maximum contact drop:

Initial: 0.150 volt.

After life test: 0.175 volt.

1/ Not applicable to -012 (table IV) nonmonitored vibration and shock.
2/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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TABLE I. Supersession data.

Superseded part number M6106/7- 1/	Superseding part number
001	MS27401-5 2/
002	MS27401-6 2/
003	MS27401-7 2/
004	MS27401-8 2/
005	No replacement
006	No replacement
007	MS27401-34 2/
008	No replacement
009	MS27401-21 2/
010	MS27401-22 2/
011	MS27401-35 2/
012	No replacement

1/ Complete part number shall consist of M6106/7- (dash number from table IV and a suffix letter M, U, X, or P to designate failure rate level). Failure rate level (percent per 10,000 cycles): M, 1.0; U, 0.5; X, 0.3; P, 0.1.

2/ Complete part number shall consist of MS27401- (dash number from above and a suffix letter M, U, X, or P designating failure rate level).

Overload current: 40 amperes dc, 60 amperes ac.

Rupture current: 50 amperes dc, 80 amperes ac.

Time-current relay characteristics at +25°C: See table II.

TABLE II. Time-current relay characteristics at +25°C. 1/ 2/

1	15A - 1 hour
2	50A - 5.0 seconds
3	100A - 1.2 seconds
4	250A - 0.2 second
5	350A - 0.1 second

1/ CAUTION: Compare with time current characteristics of the associated circuit protective device.

2/ Time-current relay characteristics at +25°C. Relays (except -005, -006, -007, -008, and -012) shall sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels for the time durations shown above. Separate relays shall be tested at 28 V dc and 115/200 V dc, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. The test shall be performed on both normally open and normally closed contacts of each relay. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at conclusion.

Loads and endurance (life): See table III.

Duty rating: Continuous.

RFI standard (applicable to coil circuits of ac operated relays): MIL-STD-461.

Part number: M6106/7- (dash number from table IV and a suffix letter M, U, X, or P designating failure rate level).

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TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main (Hz)		Aux (Hz)		Main (Hz)		Aux (Hz)	
		NO	NC	NO	NC	400	50/60	400	50/60	400	50/60	400	50/60
Resistive	100	10	10	---	---	10	---	---	---	10	---	---	---
Inductive	100	---	---	---	---	6	---	---	---	6	---	---	---
Inductive 2/	20	8	8	---	---	8	---	---	---	8	---	---	---
Motor	100	4	4	---	---	4	---	---	---	4	---	---	---
Lamp	100	2	2	---	---	2	---	---	---	2	---	---	---
Transfer, load 3/													
Mechanical life, reduced current	400	2.5	2.5	---	---	2.5	---	---	---	2.5	---	---	---
Intermediate current		Applicable per MIL-R-6106											

1/ 115/200 V ac for 60 Hz ratings. Absence of value indicates relay is not rated for 3 phase applications.
2/ M6106/7-003, -004, -005, -006, and -010 for 28 V dc inductive rating, life is 10 x 10^3 cycles.
3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplied at rating indicated.

Quality conformance inspection: Performance of groups B and C testing not applicable.

Cross-reference: See table V.

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TABLE IV. Dash numbers and operating characteristics.

Dash number 1/	Coil data: Nominal Coil	Volts	Freq Hz	Resistance Ω ±10% @ 25°C	Maximum Volts	Ampere	Max pickup voltage Normal 2/	High temp test	Cont current test	Hold voltage 2/	Dropout voltage 2/	Time (milliseconds max) Operate 3/	Release 4/	Bounce Main NO	Main NC	Aux NO	Aux NC	Terminals 5/	Mounting 5/	Maximum dimension
-001	DC	28	N/A	320	29	0.11	18	19.8	22.5	7.0	1.5	10	10	1	1	---	---	Solder hooks	RVFM	1.010
-002	DC	28	N/A	320	29	0.11	18	19.8	22.5	7.0	1.5	10	10	1	1	---	---	Plug in	"	1.010
-003	AC	115	400	N/A	122	0.04	90	95.4	103	30	5.0	15	50	1	1	---	---	Solder hooks	"	1.125
-004	AC	115	400	N/A	122	0.04	90	95.4	103	30	5.0	15	50	1	1	---	---	Plug in	"	1.125
-005	DC	48	N/A	860 (min)	50	0.08	31	34.1	39	14	2.0	15	15	1	1	---	---	Solder hooks	"	1.010
-006	DC	48	N/A	860 (min)	50	0.08	31	34.1	39	14	2.0	15	15	1	1	---	---	Plug in	"	1.010
-007	AC	115	50/60	N/A	122	0.04	95	100	105	40	5.0	20	50	1	1	---	---	Solder hooks	"	1.125
-008	AC	115	50/60	N/A	122	0.04	95	100	105	40	5.0	20	50	1	1	---	---	Plug in	"	1.125
-009	DC	28	N/A	320	29	0.11	18	19.8	22.5	7.0	1.5	10	10	1	1	---	---	Solder hooks	HFM	1.010
-010	AC	115	400	N/A	122	0.04	90	95.4	103	30	5.0	15	50	1	1	---	---	Solder hooks	"	1.125
-011	AC	115	50/400	N/A	122	0.04	90	95.4	103	30	5.0	15	50	1	1	---	---	Solder hooks	"	1.125
-012 6/	DC	28	N/A	320	29	0.11	18	19.8	22.5	7.0	1.5	10	10	1	1	---	---	Solder hooks	"	1.010

1/ Complete part number shall consist of M6106/7- (dash number from table IV and a suffix letter M, U, X, or P to designate failure rate level). Failure rate level (percent per 10,000 cycles): M, 1.0; U, 0.5; X, 0.3; P, 0.1.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ See figure 1.
6/ Vibration - No contacts monitored, level - 75 g, frequency range - 20 to 2,000 Hz. Duration - 20 minutes, 3 mutually perpendicular planes, - 1 sweep per plane, coil deenergized ascending, coil energized descending.
Shock - No contacts monitored, impact level - 2,000 g, Y2 sine wave, duration - 0.5 ms in 3 mutually perpendicular planes, 2 directions each plane, 2 shocks per direction (1 shock coil deenergized, 1 shock coil energized).

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TABLE V. Cross-reference for Government logistics support.

Superseded part M6106/7- 1/	New part number	Support with part number
001	MS27401-5 2/	MS27401-5 2/
002	MS27401-6 2/	MS27401-6 2/
003	MS27401-7 2/	MS27401-7 2/
004	MS27401-8 2/	MS27401-8 2/
005	No replacement	No replacement
006	No replacement	No replacement
007	MS27401-34 2/	MS27401-34 2/
008	No replacement	No replacement
009	MS27401-21 2/	MS27401-21 2/
010	MS27401-22 2/	MS27401-22 2/
011	MS27401-35 2/	MS27401-35 2/
012	No replacement	No replacement

1/ Complete part number shall consist of M6106/7- (dash number from table IV and a suffix letter M, U, X, or P to designate failure rate level). Failure rate level (percent per 10,000 cycles): M, 1.0; U, 0.5; X, 0.3; P, 0.1.

2/ Complete part number shall consist of MS27401- (dash number from above and a suffix letter M, U, X, or P designating failure rate level).

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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